

Learning Outcomes

At the end of the session, you will be able to:

- Write, run, and explain the difference between vector and list

Activity

1. Write a R program to store 20 records of different range of exam score. Count the number of students based on the grade as in table below. Check whether the student pass the exam (>49) or not. Return TRUE or FALSE.

Input vector: 33 24 54 94 16 89 60 6 77 61 13 44 26 24 73 73 90 39 90 54

Score	Grade	Number of students
90-100	A	
80-89	B	
70-79	C	
60-69	D	
50-59	E	
≤ 49	F	

```
score = c(33, 24, 54, 94, 16, 89, 60, 6, 77, 61, 13, 44, 26, 24, 73, 73, 90, 39, 90, 54)
print(paste("A =",length(score[score>=90&score<=100])))
print(paste("B =",length(score[score>=80&score<=89])))
print(paste("C =",length(score[score>=70&score<=79])))
print(paste("D =",length(score[score>=60&score<=69])))
print(paste("E =",length(score[score>=50&score<=59])))
print(paste("F =",length(score[score<=49])))

print(paste("Pass?",score>49))
```

2. Store the student record below in a list. Gives names to the elements in the list. Find the highest, lowest, and average of exam score.

Student Name	Exam Score
Robert	59
Hemsworth	71
Scarlett	83
Evans	68
Pratt	65
Larson	57
Holland	62
Paul	92
Simu	92
Renner	59

Sample expected outcome:

```
Highest Score:
Lowest Score:
Average Score:
Student with highest score:
Student with lowest score:
```

```
name =  
c("Robert", "Hemsworth", "Scarlett", "Evans", "Pratt", "Larson", "Holland", "Paul",  
  ", "Simu", "Renner")  
escore = c(59, 71, 83, 68, 65, 57, 62, 92, 92, 59)  
record = list(name, escore)  
names(record) <- c("StudentName", "ExamScore")  
print(record)  
print(paste("Highest Score:", max(record$ExamScore)))  
print(paste("Lowest Score:", min(record$ExamScore)))  
print(paste("Average Score:", mean(record$ExamScore)))  
print(paste("Student with highest  
score:", record$StudentName[which(record$ExamScore ==  
max(record$ExamScore))]))  
print(paste("Student with lowest  
score:", record$StudentName[which(record$ExamScore ==  
min(record$ExamScore))]))
```

3. Append the record from Question 2 to add another exam score of different subjects. Count how many students fail the Chemistry and Physics exams (≤ 49) and who got the highest/best score for both subjects (Chemistry & Physics).

Student Name	Chemistry	Physics
Robert	59	89
Hemsworth	71	86
Scarlett	83	65
Evans	68	52
Pratt	65	60
Larson	57	67
Holland	62	40
Paul	92	77
Simu	92	90
Renner	59	61

```
record[3] = list(c(89,86,65,52,60,67,40,77,90,61))  
names(record) = c("StudentName", "Chemistry", "Physics")  
print(paste("Fail  
Chemistry=", length(record$Chemistry[record$Chemistry<=49])))  
print(paste("Fail Physics=", length(record$Physics[record$Physics<=49])))  
print(paste("Highest Score:", max(record$Chemistry+record$Physics)))  
print(paste("Student with highest  
score:", record$StudentName[which(record$Chemistry==max(record$Chemistry)&  
record$Physics==max(record$Physics))]))
```

Submission

- Submit to your GA.